Product specification

Battery Model 60.8V50Ah-C103000439

IFS60.8-500-F-E3-27148127-01

Customer Name：

Manufacturer	Prepared by/date	Checked by/date	Approved by/date

Customer Approval	Checked by/date	Approved by/date

Factory：Room 301, Building 4, No.2 Xingye Road, Songshan Lake Park, Dongguan City, Guangdong Province, China

Amendment Records				
Edition	Description	Prepared by	Approved by	Date
A0	First Edition	GAO		2024-3-5

1.Scope

This specification is applied to the battery pack manufactured by Ocell.

2. Product Specification

Table 1

No.	Item	General Parameter	Remark
1	Rated Capacity	50Ah	Standard discharge 0.2C after standard charge 0.2C
2	Minimum Capacity	47.5Ah	
3	Nominal Voltage	60.8V	
4	Life Expectation	Higher than 80% of the Initial Capacity of the battery	1）Charge:CC@0.2C to 69.35V, then CV till current to 0.02C 2） Rest: 30min. 3） Discharge: 0.2C to 47.5V Temperature:20±5℃ Carry out 2000cycles
5	Discharge cut-off voltage	2.5V/Cell(≥47.5V)	
6	Charge cut-off voltage	3.65V/Cell（≤69.35V）	
7	Cell and assembly method	IFP27148127-3.2V50Ah	19S1P
8	Housing material	Iron(black)	SPCC 1.2mm

Continuous the table 1

No.	Item	General Parameter	Remark
9	Standard charge	0.2C constant current (CC) charge to 69.35V then constant voltage (CV) 69.35V charge till charge current decline to ≤0.05C	Charge time : Approx 7h
10	Standard discharge	Constant current 0.2C， Cut-off voltage 47.5V	
11	Maximum Charge Current	30A @20℃	
12	Maximum Continuous Discharge Current	50A@20℃ 60A 30S@20℃	
13	Operation Temperature Range	Charge: 0～45℃ Discharge: -20～60℃（cell surface temperature≤80℃）	60±25%R.H. 23 ± 5°C recommended
14	Storage Temperature Range	Less than 1 year : 0～25℃(SOC≥50%) Less than 3 months:-5～35℃(SOC≥50%)	60±25%R.H. at the shipment state
15	Approx. Weight	33Kg±1%	as per sample
16	Internal resistance	15~30mΩ	1KHz， AC
17	Dimension	L*W*H=(570*350*200mm ±2mm	Including handle
18	Terminals	Charge： Anderson plug (grey) 50A /600V UL3135 10AWG 230±10mm Discharge: SIN 80 (black) UL3135 6AWG 230±10mm	Positive:Red Negative:Black
19	Communication	CANbus / RS485/Bluetooth	
20	Control	Self-locking Push Switch	

3. Performance And Test Conditions

3.1 Standard Test Conditions

Test should be conducted with new batteries within one week after shipment from our factory and the batteries shall not be cycled more than five times before the test. Unless otherwise specified, test and measurement shall be done under temperature of 20±5℃ and relative humidity of 45~85%. If it is judged that the test results are not affected by such conditions, the tests may be conducted at temperature 15~30℃ and humidity 25~85%RH.

3.2 Measuring Instrument or Apparatus

3.2.1 Dimension Measuring Instrument

The dimension measurement shall be implemented by instruments with equal or more precision scale of 0.01mm.

3.2.2 Voltmeter

Standard class specified in the national standard or more sensitive class having inner impedance more than 10kΩ/V

3.2.3 Ammeter

Standard class specified in the national standard or more sensitive class. Total external resistance including
ammeter and wire is less than 0.01Ω.

3.2.4 Impedance Meter

Impedance shall be measured by a sinusoidal alternating current method (1kHz LCR meter).

3.3 Standard Charge/Discharge

3.3.1 Standard Charge : 0.2C

Charging shall consist of charging at a 0.2C constant current rate until the battery reaches 69.35V. The battery shall then be charged at constant voltage of 69.35 Volts while tapering the charge current. Charging shall be terminated when the charging current has tapered to 0.05C. Charge time: Approx 7.0h, The battery shall demonstrate no permanent degradation when charged between 0 ºC and 45 ºC.

3.3.2 Standard Discharge : 0.2C

Battery shall be discharged at a constant current of 0.2C to 47.5V @ 20±5℃

3.3.3 If no otherwise specified, the rest time between charging and discharging is 30min.

3.4 Appearance

There shall be no such defect as crack, rust, leakage, which may adversely affect commercial value of battery.

4. Handling of battery

4.1 Please read and follow the handling instructions before use. Improper use may cause heat, fire, rupture, damage or capacity deterioration of the battery. Ocell is not responsible for any accidents caused by the usage without following our handling instructions:

Warning

- Battery must be far away from heat source, high voltage, and no exposed in sunshine for long time;
- Never throw the battery into water or fire;
- Never reverse positive and negative terminals when use the battery;
- Never connect the positive and negative terminals of battery with metal;
- Never knock, throw or trample the battery;
- Never disassemble the battery without manufacturer’s permission and guidance;
- Never mixed the battery of different manufacturers, types or models with Ocell;

Tips

- Keep the battery against high temperature. Otherwise it will cause battery heat, get into fire or lose some functions and reduce the life;
- When the battery run out of power, please charge it within 15 days；
- Please use the original or recommended charger for this battery
- If the battery leaks and get into the eyes or skin, do not wipe, instead, rinse it with clean water and see doctor immediately;
- Do not use the battery when there exist abnormal conditions such as odour, discoloration, noise, leakage, serious deformation, etc.
- Please far away from children or pets.

5. Period of Warranty

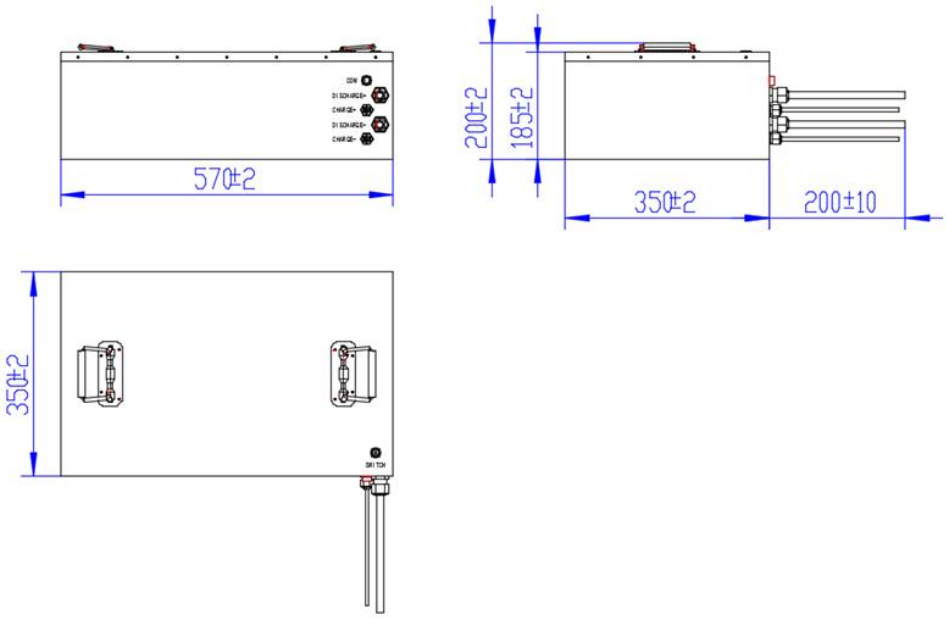
The period of warranty is 18 months from the date of shipment. Ocell guarantees to give a replacement in case of battery with defects proven due to manufacturing process instead of the customer abuse and misuse.

6. Storing the Batteries

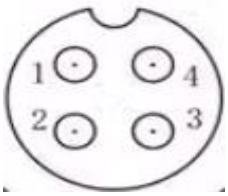
The batteries should be stored at room temperature, charged to about 30% to 50% of capacity.
We recommend that batteries be charged about once per three months to prevent over-discharge.

7. Photo:

Drawing(Unit: mm)



Port definitions



Specification	Pin	Defined declaration
GX16-4	1	CAN-H
	2	CAN-L
	3	RS485A
	4	RS485B

8. Protection circuit function

The BMS has the system self-diagnosis function. After the system is powered on, it detects the voltage, temperature, communication, internal communication and other components. Based on the battery characteristics, it can judge and alarm the battery faults such as overvoltage, undervoltage, overcurrent, overtemperature, low SOC, and consistency.

8.1 Operating Logic

8.1.1 Power on

（1）Press self-locking switch;

（2）After the system self-checks and no faults are found, the battery is powered on.

（3）Continuously send battery status messages;

8.1.2 Power off

（1） When the BMS is in working state, the self-locking switch is disconnected and the BMS cuts off the output.

8.2 Charge control

Fast charge current direction ： Charging is negative value, CAN communication baud rate: 250KB

Charging current: maximum 25A, request charging voltage is 69.35V.

① T≤0℃ request charging current 0A.

② When the battery temperature is 0℃＜T≤+50℃, the maximum allowable charging current is 25A;

③ T＞50℃, charging is prohibited;

Current reduction at the end of charging:

① When Vmax≥3550mV, request charging current to be reduced to 50% of the present current;

② When Vmax≥3600mV, request charging current to be reduced to 50% of the present current;

③ When Vmax≥3650mV, first request the charger to stop charging, output voltage and current to 0, and end charging;

④ While reducing the current, refer to the fault threshold and handling measures of the fast charging temperature sensing high temperature alarm level;

Charging request current parameter table

T \ SOC	[0%,5%)	[5%,90%)	[90%,100%)	＞100%
≤0℃	0	0	0	0
（0,5]	10	15	10	0
（5,45]	25	25	25	0
（45,50]	15	25	15	0
＞50	0	0	0	0

Fast charging high voltage power down：

Parameters	Remark	
Full charge release conditions	After SOC reaches 100%, the release condition will be activated; when Vmax≤3.35V&1S or SOC≤98%&1S is displayed, full charge is released and recharging is allowed according to the charging current limit table request	
Total charging request voltage (V)	69.35V（3.65V*19S）	Note: Full charge voltage
Fully charged release total voltage(V)	63.65V （3.35V*19S）	Note: Full charge release voltage
Charging terminal action	Request to stop charging and continue sending messages	

8.3 Discharge control

Maximum allowable discharge current（A） Discharge current direction: discharge current is positive CAN communication baud rate: 250KB

	[0%,5%)	[5%,10%)	[10%,20%)	[20%,30%)	[30%,100%]	100%
≤-20℃	0	0	0	0	0	0
（-20,0]	0	10	25	25	35	0
(0,45]	0	15	35	50	80	0
(45,55]	0	15	25	50	50	0
＞55	0	0	0	0	0	0

Voltage De-rating.

- When the single cell voltage is detected to be lower than 2.8V, it will delay for 5S and limit the maximum allowable discharge current to 25A. When the voltage recovers to 3.1V for 5S, it will recover to the previous power level.

- When the single cell voltage is detected to be lower than 2.7V, the delay is 5S, and the maximum allowable discharge current is limited to 10A. When the voltage recovers to 3.0V for 5S, it returns to the previous power level.

- When the voltage of a single cell is lower than 2.65V, the delay is 5S, and the maximum discharge current is limited to 0. When the voltage recovers to 2.95V for 5S, it returns to the previous power level.

- When the cell voltage is detected to be lower than 2.5V for 5S, the discharge MOS is cut off.

- If single cell voltage is less than 1.5V for more than 30s, Battery relay cut-off Permanently.

8.4 Sleep

1) When the protection board is in static state (no communication, no current, no balancing and overvoltage protection), it will enter the sleep state after a delay of 1 minute. After entering this state, the protection board will only reduce the detection frequency and its own power consumption. Communication, on switch, charging and discharging can automatically exit the sleep mode.

2) Standby state

-3A≤I≤3A, for 12h, cut off the charging or discharging MOS, and enter the sleep state.

9. Ocell has the right to optimize and improve the specifications on the premise of meeting customer needs.

10.specification of PCM

Cell over voltage protection	MIN	TYP	MAX	Delay time	Protection mode	After Release
Over voltage	3700mV	3750mV	3800mV	3s	The power limit current is 0，Turn off the charging MOS	
Over voltage release	3400mV	3500mV	3600mV	5s		Turn on the charging MOS
Cell under voltage protection						
Under voltage	2400mV	2500mV	2600mV	3S	The power limit current is 0，Turn off the Discharging MOS	
Under voltage release	2700mV	2800mV	2900mV	5S		Turn on the Discharging MOS
Over current (charge) protection						
1st over current (charge)	50A	55A	60A	10s	The power limit current is 0 ， Turn off the charging MOS	
1st over current (charge) release	/	0A	/	32s		Turn on the charging MOS

Over current (discharge) protection						
1st over current (discharge)	85A	90A	95A	30S	The power limit current is 0，Turn off the Discharging MOS	
1st over current (discharge) release	/	0A	/	32s		Turn on the Discharging MOS
2nd over current (discharge)	180A	190A	200A	310ms	The power limit current is 0，Turn off the Discharging MOS	
2nd over current (discharge) release	/	0A	/	32s		Turn on the Discharging MOS
Short circuit protection						
short circuit	/	330A	/	200us	The power limit current is 0，Turn off the Discharging&charging MOS	
short circuit release	/	0A	/	60s		
Over temperature (charge) protection						
Over temperature (Battery, charge)	53℃	55℃	58℃	2s	The power limit current is 0 ， Turn off the charging MOS	
Over temperature (Battery, charge) release	43℃	45℃	48℃	2S		Turn on the charging MOS
Over temperature (discharge) Protection						
Over temperature (Battery,discharge)	57℃	60℃	63℃	2s	The power limit current is 0，Turn off the Discharging MOS	
Over temperature (Battery,discharge) release	47℃	50℃	53℃	2s		Turn on the Discharging MOS
Over temperature (MOS)	85℃	90℃	95℃	2s	The power limit current is 0，Turn off the Discharging & Charging MOS	
Over temperature (MOS)release	65℃	70℃	75℃	2s		Turn on the Discharging &Charging MOS

Low temperature (charge) Protection						
Low temperature (charge)	-3℃	0℃	3℃	2s	The power limit current is 0, Turn off the charging MOS	
Low temperature (charge) release	0℃	2℃	5℃	2s		Turn on the charging MOS
Low temperature (discharge) Protection						
Low Temperature (discharge)	-23℃	-20℃	-17℃	2s	The power limit current is 0，Turn off the Discharging	
Low temperature (discharge) release	-17℃	-15℃	-13℃	2s		Turn on the Discharging MOS
Balancing function						
Cell Balance Threshold	3250mV	3300mV	3350mV			
Cell Balance Min	/	20mV				
Cell Balance current	50mA		200mA			
Cell Balance Interval	/					